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# MScFE 632: Machine Learning in Finance GWP1
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# Step 1: Category 2: k-means clustering Computation
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt

# 1. Load dataset
df = pd.read_csv('s_and_p_500_closing.csv')

# 2. Clean data
if 'Date' in df.columns:
    df = df.drop(columns=['Date'])
df = df.apply(pd.to_numeric, errors='coerce')
df = df.fillna(method='ffill').dropna()

# 3. Compute returns
returns = df.pct_change().dropna()

# 4. Stock-level features
features = pd.DataFrame({
    "Returns": returns.mean(),
    "Volatility": returns.std()
})

features.dropna(inplace=True)

# 5. Standardize
scaler = StandardScaler()
X = scaler.fit_transform(features)

# 6. Elbow method
distortions = []
K_range = range(2, 10)
for k in K_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X)
    distortions.append(km.inertia_)
plt.figure()
plt.plot(list(K_range), distortions)
plt.title("Elbow Curve (SSE vs K)")
plt.xlabel("Number of clusters (K)")
plt.ylabel("SSE (Inertia)")
plt.show()

# 7. Fit final K-Means
best_k = 5
kmeans = KMeans(n_clusters=best_k, random_state=42, n_init=10)
features["Cluster"] = kmeans.fit_predict(X)

# 8. Visualization
plt.figure()

for c in range(best_k):
    cluster_data = features[features["Cluster"] == c]
    plt.scatter(cluster_data["Returns"], cluster_data["Volatility"], label=f"Cluster {c}")

centroids = kmeans.cluster_centers_
centroids = scaler.inverse_transform(centroids)
plt.scatter(centroids[:, 0], centroids[:, 1], s=200, c="black", marker="X", label="Centroids")
plt.title("Stock Clusters (K-Means)")
plt.xlabel("Mean Return")
plt.ylabel("Volatility")
plt.legend()
plt.show()

# 9. Results
print("\nCluster Summary:")
print(features.groupby("Cluster").mean())

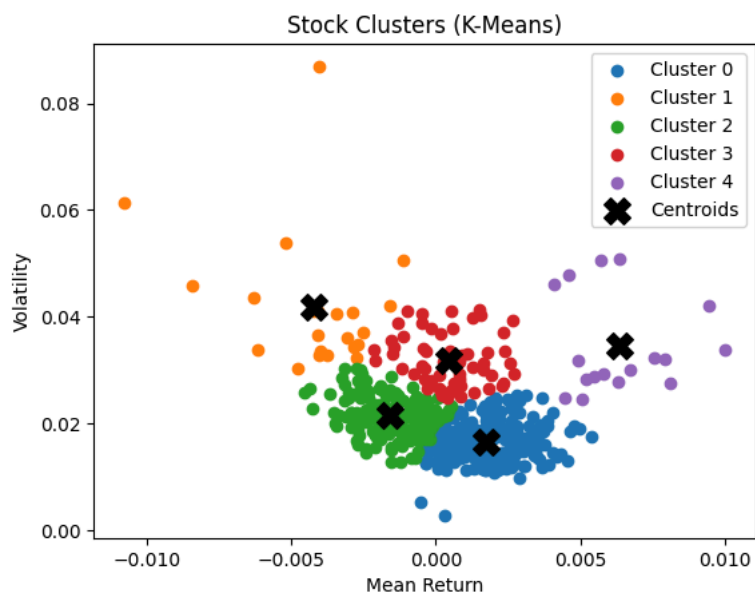
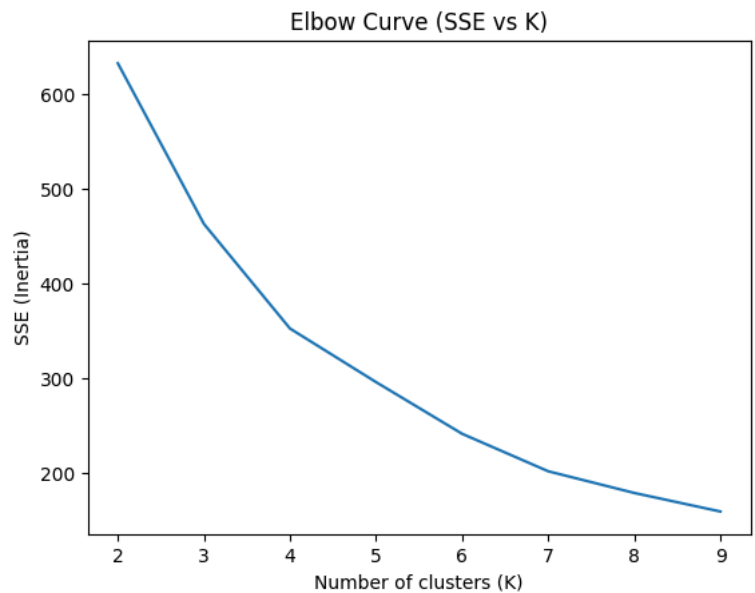
print("\nStocks per cluster:")
print(features["Cluster"].value_counts())

# Optional: stock assignment list
print("\nStock assignments:")
for stock, cluster in features["Cluster"].items():

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for stock, cluster in reduced_cluster_items():  
    print(stock, cluster)
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/tmp/ipykernel_17/3554583010.py:20: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future
df = df.fillna(method='ffill').dropna()



Cluster Summary:

	Returns	Volatility
Cluster		
0	0.001737	0.016604
1	-0.004221	0.041974
2	-0.001586	0.021535
3	0.000447	0.031838
4	0.006334	0.034666

Stocks per cluster:

Cluster	
0	230
2	174
3	61
1	22
4	17

Name: count, dtype: int64

Stock assignments:

A 2
AAL 3
AAP 2
AAPL 0
ABBV 0
ABC 0
ABMD 3
ABT 2
ACN 2
ADBE 2
ADI 3
ADM 0
ADP 0
ADSK 2
AEE 0
AFP 0

AES 0
AFL 0
AIG 0
AIZ 0
AJG 0
AKAM 0
ALB 3
ALGN 3
ALK 3
ALL 0
ALLE 2
AMAT 2
AMCR 0
AMD 1
AME 2
AMGN 0
AMP 2
AMT 0
AMZN 3
ANET 3
ANSS 2
ANTM 0
AON 0
AOS 2
APA 4
APD 2
APH 2
APTV 1
ARE 0
ATO 0
ATVI 0
AVB 0
AVGO 0
AVY 2
AWK 0
AXP 0
AZO 0
BA 2
BAC 2
BAX 2
BBWI 3
BBY 3
BDX 0
BEN 2
BF-B 0
BIIB 2
BIO 2
BK 2
BKNG 3
BKR 4
BLK 2
BLL 2
BMY 0
BR 2
BRK-B 0
BRO 0
BSX 0
BWA 2
BXP 0
C 2
CAG 0
CAH 0
CARR 2
CAT 0
CB 0
CBOE 0
CBRE 2
CCI 0
CCL 3
CDAY 1
CDNS 3
CDW 2
CE 2
CEG 4
CERN 0
CF 4
CFG 2
CHD 0
CHRW 2
CHTR 2
CI 0
CINF 0
CL 0
CLX 2
CMA 2
CMCSA 2
CME 0
CMG 3
CMI 2
CMS 0

CNC 0
CNP 0
COF 2
COO 0
COP 0
COST 0
CPB 0
CPRT 2
CPT 0
CRL 2
CRM 2
CSCO 2
CSX 0
CTAS 0
CTLT 2
CTRA 4
CTSH 0
CTVA 0
CTXS 0
CVS 0
CVX 0
CZR 3
D 0
DAL 3
DD 2
DE 0
DFS 3
DG 0
DGX 0
DHI 2
DHR 2
DIS 2
DISH 3
DLR 0
DLTR 0
DOV 2
DOW 0
DPZ 2
DRE 0
DRI 2
DTE 0
DUK 0
DVA 0
DVN 4
DXC 3
DXCM 3
EA 2
EBAY 2
ECL 2
ED 0
EFX 2
EIX 0
EL 2
EMN 2
EMR 0
ENPH 4
EOG 0
EPAM 1
EQIX 0
EQR 0
ES 0
ESS 0
ETN 2
ETR 0
ETSY 1
EVRG 0
EW 0
EXC 0
EXPD 2
EXPE 3
EXR 0
F 1
FANG 3
FAST 0
FB 1
FBHS 2
FCX 3
FDS 0
FDX 2
FE 0
FFIV 2
FIS 2
FISV 2
FITB 2
FLT 0
FMC 0
FOX 0
FOXA 0
FRC 2
FRT 0
FTNT 3